Reproducable Workflows

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«FAIR-by-design»

Programme	Call ID	Full call title	Total topic budget	Indicative EU funding per project
2026	HORIZON-CL6-2026-01-BIODIV-01	Understanding and tackling the decline of insects	€13 million	~€6.5 million; 2 projects FFG
2026	HORIZON-CL6-2026-01-BIODIV-02	Developing methods to assess the presence, functions and sensitivity of groundwater ecosystems	€10 million	~€5 million; 2 projects FFG
2027	HORIZON-NATURE-2027-01-01	Understanding the nature and the extent of the effects of PFAS on biodiversity and ecosystem services	€11.45 million	~€5.7 million; 2 projects 2027 +1
2027	HORIZON-NATURE-2027-02-01	Assessing direct and indirect drivers of biodiversity decline of invertebrates	€15 million	~€7.5 million; 2 projects FFG
2027	HORIZON-NATURE-2027-02-02	Living labs co-creating solutions for the restoration of ecosystems addressing implementation bottlenecks and improving resilience to risks	€60 million	~€10 million; 6 projects European Commiss
2027	HORIZON-CL6-2027-01-BIODIV-04	Living Labs for the eradication and/or management of invasive alien species	€13 million	~€13 million; 1 project European Rese +1
2027	HORIZON-NATURE-2027-02-05	Cost-effective biodiversity monitoring for EU policies	€16 million	€5–6 million; 3 projects FFG
2027	HORIZON-MISS-2027-05-SOIL-04-two-stage	Participatory research on the health of communities in contact with polluted soils	€11 million	Check topic conditions for the expected contribution Horizon-europe.gc

This is an AI-created summary of the current (and recently closed) HORIZON Europe calls related to ecology that explicitly ask for «FAIR-by-design» workflows

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FAIR by Design Data Workflows for Ecology

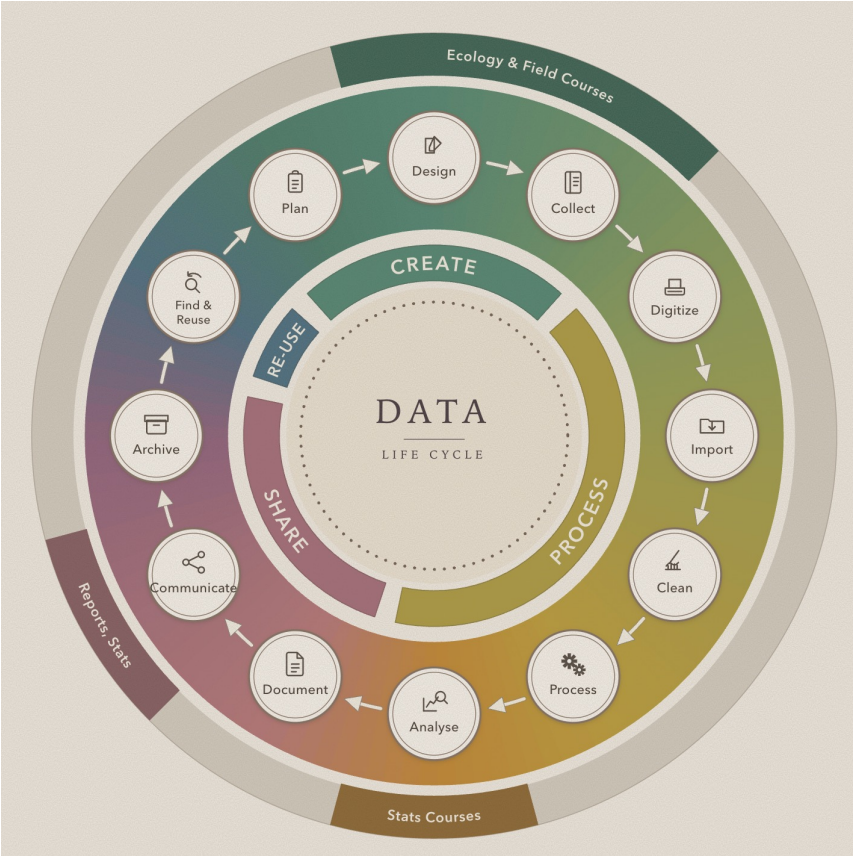
Making ecological data Findable, Accessible, Interoperable, and Reusable from the start

- From field observations to reusable research data
- Embedding good data practices throughout the research lifecycle
- Building reproducible, transparent, and collaborative ecological science

Ecological research increasingly depends on combining data across sites, species, timescales, and disciplines. The challenge is not simply collecting more data, but ensuring that data remain understandable and useful beyond the original project.

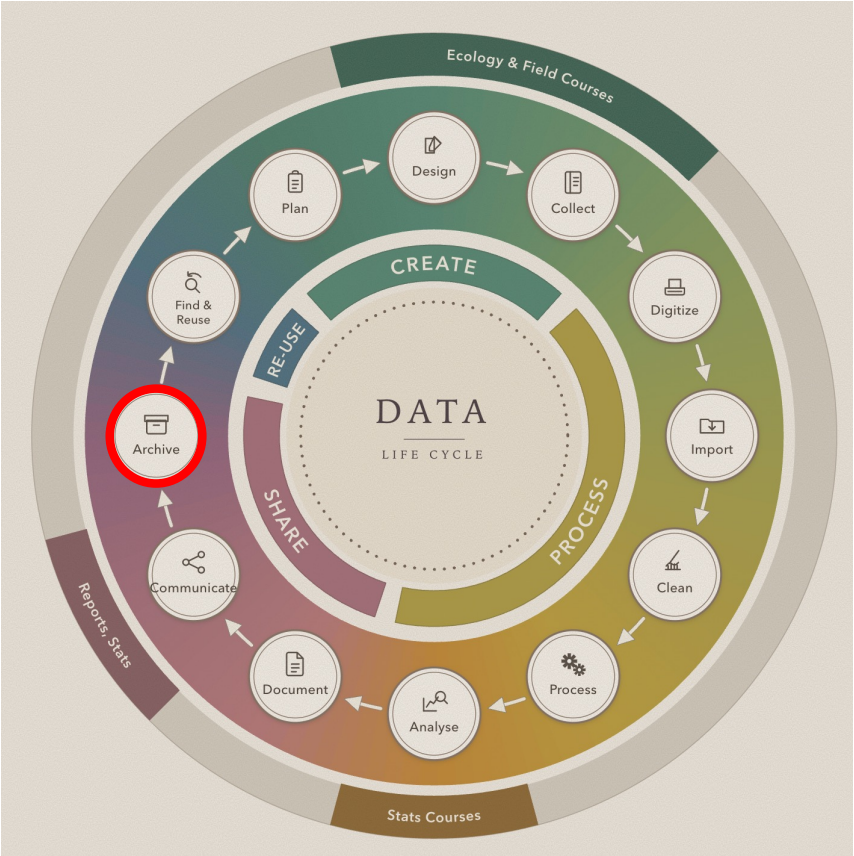
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Data Workflows



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Data Workflows



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Why Ecological Data Workflows Matter

- Ecology generates diverse data: field observations, experiments, sensor measurements, images, DNA sequences, and remote sensing products.
- Data are often collected by multiple people, across locations and over long periods.
- Inconsistent formats, missing metadata, and undocumented decisions make integration difficult.
- Research projects may end, but the scientific value of their data can continue to grow.
- A well-designed workflow protects that value.

Ecological datasets often outlive the projects, funding, software, and people that created them. Workflow design helps preserve both the data and the knowledge needed to interpret them.

FAIR Principles

F — **Findable**

Data and metadata can be discovered and identified reliably.

A — **Accessible**

People and systems know how to retrieve data under clearly defined conditions.

I — **Interoperable**

Data can be combined with other data using shared formats, vocabularies, and standards.

R — **Reusable**

Data have sufficient context, provenance, and licensing information to support new uses.

FAIR Is Not the Same as Open Data

- FAIR is about usability. Can data be found, accessed appropriately, understood, and reused?
- Open is about permissions. Can anyone access and use the data without restrictions?
- Sensitive ecological data may need controlled access.
- Data involving threatened species, private land, Indigenous knowledge, or personal information may require safeguards.
- Metadata can often be public even when the underlying data are restricted.

Make data as open as possible and as restricted as necessary.

Start with FAIR by Design

At the planning stage:

- Identify the research questions and expected data products.
- Agree on data formats, variable names, units, and metadata.
- Select suitable identifiers, vocabularies, and classification systems.
- Decide how data quality will be assessed.
- Plan storage, version control, backup, access, and long-term preservation.
- Assign responsibilities for documentation and publication.

Practical example: Before a multi-site vegetation survey, agree on plot identifiers, species naming conventions, coordinate reference systems, sampling dates, and protocols for recording missing observations.

F: Make Ecological Data Findable

Findability starts with discoverable metadata and persistent identification.

- Assign a persistent identifier, such as a DOI, to a published dataset where appropriate.
- Provide a descriptive, informative title.
- Record authors, affiliations, publication date, and responsible organisations.
- Add keywords, taxonomic coverage, geographic coverage, and temporal coverage.
- Describe the research context, sampling methods, and relevant environmental variables.
- Register metadata in a repository or catalogue that others can search

A: Make Data Accessible

Accessibility means that users can determine how to retrieve data and under what conditions.

- Deposit data in a trusted, appropriate repository.
- Use standard communication protocols, typically HTTPS.
- Provide clear access instructions and stable landing pages.
- Document licensing, embargoes, and any restrictions.
- Keep metadata available even if data access is restricted.
- Explain how to request access to sensitive or controlled datasets.

Ecological considerations

- Exact locations of endangered species may need to be withheld or generalised.
- Landowner agreements may limit redistribution.
- Human-related data may require privacy protections and controlled access

I: Make Data Interoperable

Interoperability enables datasets from different sources to work together.

- Use documented, widely supported formats such as CSV, GeoPackage, GeoTIFF, or NetCDF, as appropriate.
- Adopt shared vocabularies and recognised domain standards.
- Use consistent units and coordinate reference systems.
- Distinguish missing, not applicable, and zero values.
- Use stable identifiers for taxa, locations, sampling events, and instruments where possible.
- Document mappings when converting between different schemas.

Example: Two vegetation surveys can only be reliably compared if their species names, sampling units, plot definitions, dates, and measurement methods are sufficiently compatible or their differences are explicitly described.

R: Make Data Reusable

Reuse requires enough information to interpret data correctly and assess whether they are suitable for a new purpose.

Include:

- Detailed sampling and measurement methods.
- Definitions of variables, units, and codes.
- Data quality checks and known limitations.
- Information on missing values and exclusions.
- Processing history and links to analytical code.
- Clear usage licences and citation instructions.

The reuse test: Could a researcher outside your team understand the dataset, evaluate its limitations, cite it, and use it without contacting the original data collector?

Metadata: The Context That Makes Data Useful

For ecological research, useful metadata include:

- Study: research objectives, study design, and project description.
- Sampling: methods, sampling effort, frequency, and protocols.
- Space: coordinates, coordinate reference system, spatial precision, and site descriptions.
- Time: observation dates, time zones, and temporal resolution.
- Variables: definitions, units, data types, and permitted values.
- Provenance: collectors, instruments, processing steps, and version history.

Example: A temperature value of `18.4` is difficult to interpret without knowing whether it is measured in °C, whether it refers to air or soil, and when and where it was measured.

Ecological Data Standards in Practice

Standard or resource	Typical application
Darwin Core	Biodiversity occurrences, taxa, and related information
Ecological Metadata Language (EML)	Ecological datasets and research context
ABCD	Biodiversity data exchange, especially specimen and observation records
ISO 19115	Geographic information metadata
MIxS	Minimum information about biological and environmental sequence data
OBIS/GBIF data practices	Sharing marine biodiversity and broader biodiversity occurrence data

- Choose standards according to the data type and intended community.
- Use existing standards rather than inventing new structures unnecessarily.
- Document any extensions, local conventions, or transformations.
- Check repository requirements before finalising the workflow

Design Data Collection for Quality

Good downstream data depend on good upstream collection.

Build quality assurance into field and laboratory procedures:

- Use standardised digital forms where appropriate.
- Validate values at entry—for example, dates, units, and allowed categories.
- Require essential identifiers and record who collected each observation.
- Synchronise taxonomic names and controlled vocabularies.
- Record instrument calibration, sampling effort, and protocol deviations.
- Preserve original observations separately from corrected or transformed data.

Example: A field survey app

A species observation form can require a plot ID, date, observer, and taxon, while flagging impossible coordinates or unexpected measurement ranges.

Make Data Cleaning and Transformation Reproducible

A robust workflow separates:

- Raw data: the original, preserved observations.
- Validated data: records checked against predefined rules.
- Harmonised data: standardised names, units, formats, and identifiers.
- Analysis-ready data: derived variables and datasets prepared for modelling.
- Published data: documented, quality-checked outputs suitable for sharing.

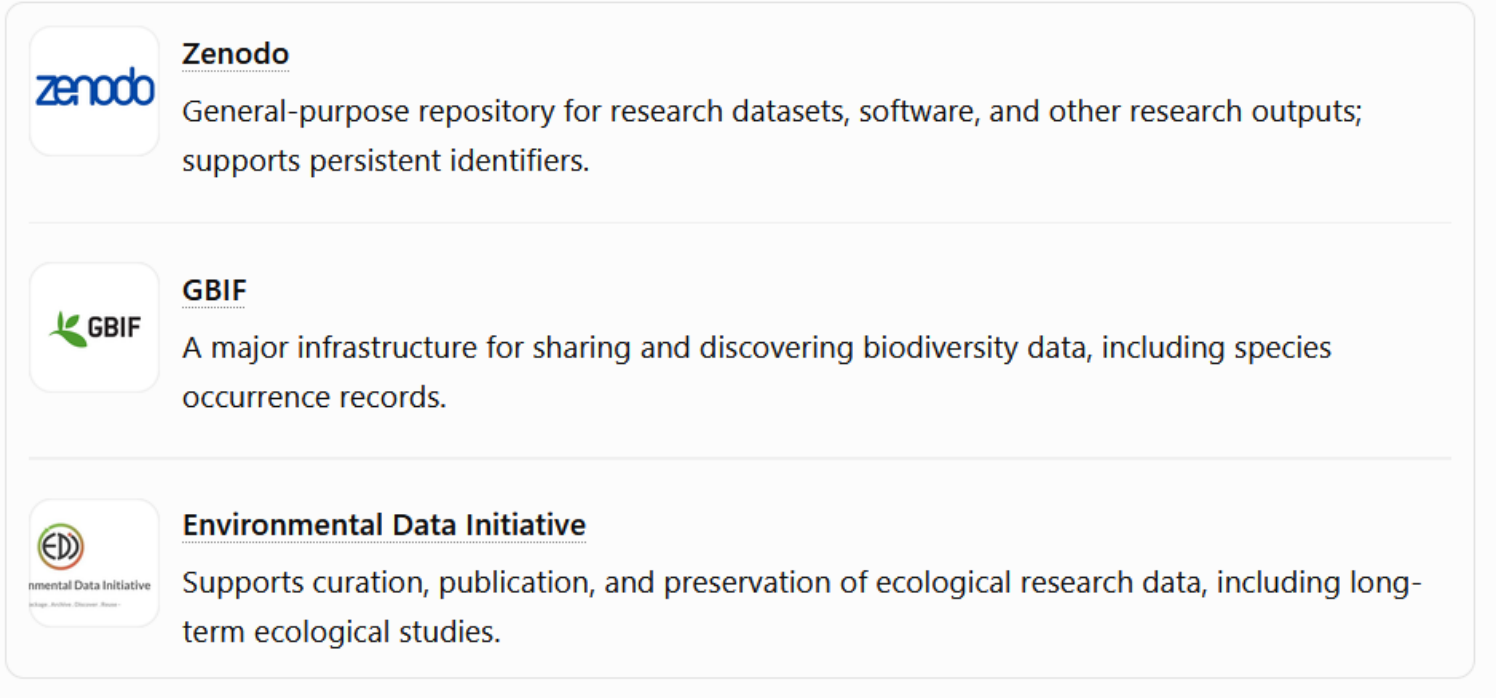
- Use scripts for repeatable transformations whenever feasible.
- Record corrections, exclusions, and assumptions.
- Keep intermediate files and outputs clearly labelled.
- Make the workflow rerunnable from documented inputs.
- Avoid undocumented manual edits to final spreadsheets.

Reproducible Analysis: Code, Environments, and Provenance

- FAIR data and reproducible analysis reinforce one another.
- Store analysis scripts in version control, for example with Git.
- Use meaningful commits and release tags.
- Record software versions, dependencies, and computational requirements.
- Use workflow tools where useful to manage dependencies and rerun analyses.
- Link outputs to their inputs, code, and processing history.
- Archive a citable release of the code alongside the published research.

Possible tools: R, Python, Quarto, Jupyter, Git, renv, conda, Snakemake, and targets.

From Local Files to Trusted Repositories



Choose a repository based on: disciplinary fit, metadata support, persistent identifiers, licensing, preservation policy, access controls, and community practices.

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A Worked Example: A Multi-Site Plant Diversity Study

Imagine a team studying how plant diversity changes along an elevation gradient.

Without a FAIR-by-design workflow:

- Each site uses a different spreadsheet structure.
- Species names and units vary between teams.
- Plot IDs are duplicated or ambiguous.
- Cleaning decisions are undocumented.
- Combining data requires extensive manual work.

With a FAIR-by-design workflow:

- Plot and sampling-event identifiers are defined before fieldwork.
- Digital forms apply shared vocabularies and validation rules.
- Taxonomic names and measurement units are harmonised.
- Raw observations are preserved; transformations are scripted.
- Metadata document sites, sampling methods, temporal coverage, and limitations.
- Data and code are published with clear licences and persistent identifiers.

Common Barriers and Practical Solutions

Barrier	Practical response
Limited time	Use templates and automate routine metadata capture.
Inconsistent field practices	Agree on shared protocols and identifiers before collection.
Legacy datasets	Prioritise high-value datasets and document known gaps.
Sensitive locations	Generalise coordinates or provide controlled access where appropriate.
Complex standards	Adopt a fit-for-purpose standard and expand incrementally.
Unclear responsibilities	Assign ownership for validation, documentation, and publication.
Fear of losing control	Establish clear licences, citation practices, and contributor roles.

Make FAIR the Default, Not the Final Step

- FAIR practices improve data quality, collaboration, reproducibility, and long-term scientific value.
- The most effective workflows embed metadata, standards, validation, and provenance from the beginning.
- Tools help, but clear protocols, shared responsibilities, and good research culture are equally important.
- FAIR implementation can be incremental: start with the most valuable and achievable improvements.

Every ecological dataset is an investment in future discovery. Designing for FAIR from the start helps ensure that investment can be found, understood, combined, and reused.